Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

ANNUAL REVIEWS

Annual Review of Animal Biosciences

Adipose Tissue Inflammation: Linking Physiological Stressors to Disease Susceptibility

Barry J. Bradford[1] and G. Andres Contreras[2]

[1]Department of Animal Science, College of Agriculture and Natural Resources, Michigan State University, East Lansing, Michigan, USA; email: bjbrad@msu.edu

[2]Department of Large Animal Clinical Sciences, College of Veterinary Medicine, Michigan State University, East Lansing, Michigan, USA; email: contre28@msu.edu

ANNUAL REVIEWS CONNECT

www.annualreviews.org

- Download figures
- Navigate cited references
- Keyword search
- Explore related articles
- Share via email or social media

Annu. Rev. Anim. Biosci. 2024. 12:261–81

First published as a Review in Advance on December 8, 2023

The *Annual Review of Animal Biosciences* is online at animal.annualreviews.org

https://doi.org/10.1146/annurev-animal-021122-113212

Copyright © 2024 by the author(s). This work is licensed under a Creative Commons Attribution 4.0 International License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See credit lines of images or other third-party material in this article for license information.

Keywords

immunity, metabolism, homeostasis, programming, inflammation, endocrinology

Abstract

The study of adipose tissue (AT) is enjoying a renaissance. White, brown, and beige adipocytes are being investigated in adult animals, and the critical roles of small depots like perivascular AT are becoming clear. But the most profound revision of the AT dogma has been its cellular composition and regulation. Single-cell transcriptomic studies revealed that adipocytes comprise well under 50% of the cells in white AT, and a substantial portion of the rest are immune cells. Altering the function of AT resident leukocytes can induce or correct metabolic syndrome and, more surprisingly, alter adaptive immune responses to infection. Although the field is dominated by obesity research, conditions such as rapid lipolysis, infection, and heat stress impact AT immune dynamics as well. Recent findings in rodents lead to critical questions that should be explored in domestic livestock as potential avenues for improved animal resilience to stressors, particularly as animals age.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

INTRODUCTION

The adipose tissue (AT) is a multisite organ that serves as the major store of energy reserves (~80%) in large mammals, a thermogenic regulator, and an endocrine tissue that modulates metabolism and immune function (1, 2). AT integrates metabolic and inflammatory signals through autocrine, paracrine, and endocrine cues using lipids (e.g., oxylipins and ceramides) and proteins (e.g., adipokines, chemokines) as messengers. AT maintains lipid energy reserves in a dynamic process that continuously releases (i.e., lipolysis) and synthesizes and esterifies fatty acids (i.e., lipogenesis) in response to circadian energy fluxes (3). Furthermore, AT supports immune function by providing energy for immune responses and by serving as a reservoir for numerous immune cell populations. During metabolic stress (e.g., negative energy balance) and metabolic and infectious diseases, inflammation develops in AT as an adaptation to alterations driven by excessive lipolysis, lipogenesis, or pathogen infections. In certain diseases, AT inflammation is a determinant pathogenic factor; in others, it is a consequence. We provide an overview of AT inflammatory responses during metabolic stress and metabolic and infectious diseases and identify knowledge gaps that need to be addressed to improve our understanding of AT biology in health and disease in production animals.

ADIPOSE TISSUE DEPOTS' STRATEGIC LOCATION: EVERYWHERE

AT depot sites are broadly classified as visceral AT (VAT) and subcutaneous AT (SCAT). Visceral locations include depots in the thoracic and abdominal cavities, whereas subcutaneous depots are heterogeneous fat pads localized under the dermal layer. However, both visceral and subcutaneous depots are highly diverse, and their biology is deeply interdependent with the function of the organs they surround (4). For example, perirenal and renal sinus VAT modulates kidney function, and dysregulated inflammatory responses in this pad are part of the pathogenesis of kidney disease (5, 6). Perivascular AT is a VAT surrounding vessels larger than 200 μm in diameter, and its metabolites are potent vasoactive compounds (7). Periovarian fat exerts paracrine control of folliculogenesis, and excessive inflammation and expansion of the depot lead to reproductive failure in monogastric species (8). On the other hand, available evidence points to a potent role for chicken VAT in regulation of reproduction (9); because of this and the limited data available on avian AT regulatory function, this review focuses on mammalian species.

Independent of anatomical location, AT depots are defined by the presence of adipocytes embedded in an extracellular matrix structure accompanied by a stromal vascular fraction (SVF; non-buoyant cells) that includes immune cells, fibroblasts, vascular cells (endothelial and smooth muscle cells), multipotent stem cells, and adipocyte progenitors. There are three major adipocyte phenotypes: brown, white, and beige. Brown adipocytes are thermogenic cells enriched with numerous mitochondria and lipid droplets (multilocular). White adipocytes are specialized in neutral lipid storage and contain a single large lipid droplet (unilocular). Beige fat cells are intermediate adipocytes with the capacity to store lipids and generate heat. Independent of the phenotype, adipocytes originate from precursors in the neural crest and mesoderm (10).

Adipose Tissue Development: Effects on Adipose Tissue Function and Inflammatory Responses

In farm animals, the development of AT depots begins at the end of the first third of gestation [reviewed in detail by Louveau et al. (11)]. In pig fetuses, the subcutaneous AT of the backfat area emerges first at 45 days of gestation, and by 1 day postnatal, all subcutaneous AT depots are populated with multilocular adipocytes (12, 13). As for visceral AT depots, pigs' perirenal

fat becomes visible first at 70 days of gestation, whereas in cattle, it develops around 80 days of gestation (14, 15). Toward the end of the third trimester, both subcutaneous and omental AT are present in calves, and the newborn calf has adipocytes in all anatomical AT sites (16). In sheep, AT depots appear first in the perirenal area during the second third of gestation (day 70 of 147 days of gestation). Newborn lambs have both subcutaneous and visceral AT depots, with perirenal being the most developed site (17).

Ruminants are born with fully developed AT depots, although with minimal lipid storage. Rapid expansion of AT mass occurs during the nursing and prepubertal stages, as fatty acid and triglyceride synthesis fill up adipocyte lipid droplets. Preweaning AT growth appears to be explained by adipocyte hypertrophy, as the number of adipocytes remains relatively stable (18). During these stages of development, ruminants already show a marked preference for acetate and lactate versus glucose as substrates for de novo lipogenesis (i.e., fatty acid synthesis) in adipocytes (19).

During gestation and early postnatal development, periods of acute negative energy balance, the occurrence of metabolic and infectious diseases, and exposure to extreme environmental conditions impact AT development and may predispose the offspring to inflammatory and metabolic diseases. For example, energy restriction in gestating ewes during the first and second third of gestation increases AT mass in their lambs compared to the offspring of ewes fed to meet energy requirements, despite similar fetal weights (20). The underlying mechanisms for this response are complex and involve endocrine and metabolic changes in both the mother and offspring. Reduced maternal secretion of leptin throughout gestation and excessive proliferation and lipogenic capacity of fetal adipocyte progenitors are among the factors contributing to enhanced visceral adiposity in the offspring (21, 22). In ruminants, rodents, and humans, the offspring from mothers with fetal growth restriction have increased visceral adiposity with increased infiltration of immune cells in the omental and mesenteric AT depots during adulthood (21, 23). A similar response is observed in neonates that experience early life stress, such as maternal separation and other traumatic events (24, 25). These responses to prenatal and neonatal alterations in AT development may compound the complexity of progeny's AT responses during metabolic and infectious challenges in adulthood.

Adipose Tissue Cellular Composition

Far from being the passive energy bank that it was once characterized as, AT is a dynamic tissue type containing a variety of cellular components in constant communication. In addition to mature adipocytes and progenitor cells, white AT contains endothelial, mesenchymal, mesothelial, smooth muscle, myeloid, and lymphoid cells (26). Although mature, lipid-filled adipocytes are very large and therefore comprise the majority of AT volume in a well-nourished animal, adipocytes in fact comprise only 8–30% of the cells in mouse and human AT (27) and approximately 40% of cells in bovine AT (26).

The number of adipocytes remains relatively constant in adult mammals, but the adipocyte turnover rate varies by species. In rodents, adipocytes are replenished at a rate of 1–5% daily (28), whereas in humans, the annual death rate of fat cells is 8–12% (29). The rate of replacement in ruminants and swine is unknown. However, similar to rodents, it is expected to be dynamic, especially during periods of rapid growth and following negative energy balance periods. To support turnover rates, AT possess a population of adipose stem and progenitor cells comprising 15–40% of cells, providing for the remarkable plasticity of the tissue, whereas endothelial cells account for another 5–10% of cells. Visceral AT consistently contains mesothelial cells (10–40% of cells), which are not detected in subcutaneous depots (26, 27).

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

Adipose tissue macrophages. The AT has all the cellular components required to mount an effective immune response. Macrophages are the most abundant immune cells, comprising 6–15% of cells in healthy animals (30) and upward of 50% of AT cells in perigonadal depots of obese mice. In dairy cattle, AT macrophages (ATMs) in visceral depots represent 6–13% of cells in the SVF (31). In cows with metabolic diseases such as displaced abomasum and clinical ketosis, ATMs make up 20% of the SVF and are potent drivers of the inflammatory response observed in these diseases (32). ATMs can be either tissue-resident (yolk sac and fetal liver origin) or recruited monocyte-derived macrophages (33). Independent of their origin, ATMs develop an immune phenotype that was initially described in the spectrum of M1 (classical or pro-inflammatory) to M2 (alternative or pro-resolving). However, proteomic and transcriptomic studies at the tissue and single-cell level in rodents (34, 35) identify a variety of ATM subsets that do not fit the M1/M2 range and rather are defined by their metabolic function. In dairy cattle, using single-nucleus RNA sequencing, Michelotti et al. (26) determined that macrophages represent 80% of immune cells in AT and identified five subtypes of macrophages. These populations included M1- and M2-like phenotypes but also two subtypes of lipid metabolism–associated macrophages. The latter are key regulators of AT lipid metabolism in health and in obesity and metabolic syndrome in humans (33). Taken together, these findings from omics studies highlight the need for future research on ATM biology in production animals, especially those that face cyclical metabolic challenges, such as dairy species and sows.

Other immune cells. Other cells of the innate immune system are also residents of the AT. Neutrophils constitute ~1% of the SVF cells in monogastrics and ruminants (36). In cattle, SCAT and VAT harbor limited numbers (<0.1% of SVF) of basophils and eosinophils (36). The AT hosts both T and natural killer (NK) lymphocytes. In dairy cattle, T cells comprise ~3% and NK ~2% of SVF cells in VAT and SCAT. Remarkably, 40–45% of T cells in bovine AT are of the γδ phenotype (37); this observation coincides with AT's role as a memory T-cell reservoir as described in rodents and humans (38).

ROLE OF ADIPOSE TISSUE IN SYSTEMIC PHYSIOLOGY

Given that AT contains most of the labile energy stores in a well-nourished animal, its role in meeting basic metabolic energy requirements during periods of dietary inadequacy is clear. In fasted animals, lipolysis of AT triglyceride rapidly releases free fatty acids (FFAs) (39) that can be directly oxidized by many tissues. Nerve tissue cannot rely primarily on fatty acid oxidation for ATP production, but conversion of fatty acids to ketone bodies by the liver provides an oxidizable substrate to maintain these cell types when glucose availability dips. In addition, glycerol released as another product of lipolysis can serve as a key substrate for liver gluconeogenesis in the fasted state.

More recent work has shed new light on the importance of AT for nutrient buffering in the fed state as well. When excess nutrients are available, endocrine signals such as insulin trigger anabolic processes in AT that drive uptake of metabolites such as glucose and FFAs, resulting in (depending on the species) de novo fatty acid synthesis and triglyceride synthesis. In both lipodystrophy and severe obesity, a lack of capacity for additional triglyceride storage ultimately results in diminished nutrient uptake by AT, leading to glucose intolerance and hyperlipidemia (40). This reduction of homeostatic buffering capacity in AT has serious implications for systemic metabolic health, contributing to the development of diabetes, fatty liver, and other obesity comorbidities. Although obesity is not defined in livestock, breeds with genetic propensity for high adiposity develop all hallmarks of metabolic syndrome when provided excess energy for sustained periods of time (41) (see the sidebar titled Ectopic Fat).

ECTOPIC FAT

Another implication of dysregulated systemic lipid metabolism is ectopic fat deposition, which is defined broadly as development of lipid droplets at anatomical sites where they are not typically found. Most often, ectopic fat is in the form of intracellular lipid droplets in parenchymal cells, which may impair cellular function. Fatty liver is among the most widely studied forms of ectopic fat deposition and is a significant health risk for humans and dairy cattle (135, 136), but lipid accumulation likely occurs simultaneously in multiple organs in some situations, impacting several physiological systems (137). Skeletal muscle displays some stark species differences in terms of lipid accumulation. Whereas in humans accumulation of fat within the muscle fiber is primarily intramyocellular lipid, in livestock species, intramuscular adipocytes develop (referred to as marbling in meat), with little apparent intramyocellular lipid accumulation (138).

Regulatory Roles

The mechanisms above focus on the systemic importance of AT serving as a nutrient bank, preventing both toxicity from excessive circulating nutrients and starvation during extended periods of nutrient deficiency. However, the role of AT in metabolic homeostasis does not end there. As Gordon Kennedy (42) famously predicted more than 70 years ago, leptin was the first hormone identified that is produced by adipocytes (43). Since that time, the adipokine class of hormones has exploded, with dozens of factors produced by AT that can affect other organs.

Figure 1 points to a handful of adipokines that contribute to regulation of systemic metabolism along either anabolic/catabolic or inflammatory/anti-inflammatory axes. However, these distinctions are actually quite blurry. For example, although adipsin serves an anabolic role by promoting pancreatic beta cell stability and subsequently insulin action (44), it is also known as factor D of the complement cascade, promoting inflammation and other aspects of an immune response (45).

These overlapping metabolic and inflammatory roles are perhaps no accident. Meizlish et al. (46) presented a compelling framework for understanding inflammation as a suprahomeostatic regulatory tool. In this paradigm, classical endocrine axes provide adequate control circuits to maintain homeostasis in the face of normal variability in the environment facing an animal. However, in scenarios where illness, for example, necessitates a much more dramatic response to restore homeostasis, inflammatory circuits are needed because their impacts on cellular physiology are more extreme or work in concert with classical endocrine axes to amplify or diminish their effects.

One example that may shed light on this paradigm is the impact of infection-induced cytokine signaling on circulating insulin and glucose. Maintaining glucose availability is critical to host survival in severe viral infections (47), likely because it is a required substrate for activated immune cells. Glucose can be preserved by reducing circulating insulin concentrations (classical endocrine control); however, insulin also enhances lymphocyte function (48), making this solution suboptimal during disease. In this scenario, inflammatory signaling helps to protect homeostasis by inducing insulin resistance in key tissue types (e.g., muscle and adipose tissue), minimizing flux of glucose to these tissues despite elevated insulin concentrations. This control network therefore allows both preservation of glucose and enhanced insulin concentrations at the same time, to promote pathogen clearance and a resumption of normal physiology (49).

Importantly, this suprahomeostatic role for physiological inflammation may be employed in response to a far wider variety of stressors than infectious disease, potentially including extreme nutrient imbalance (excess or deficiency), hypoxia, tissue trauma, and heat stress. Whether and how AT contributes to the systemic inflammatory and endocrine responses to physiological stressors is a key focus of the remainder of this review.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

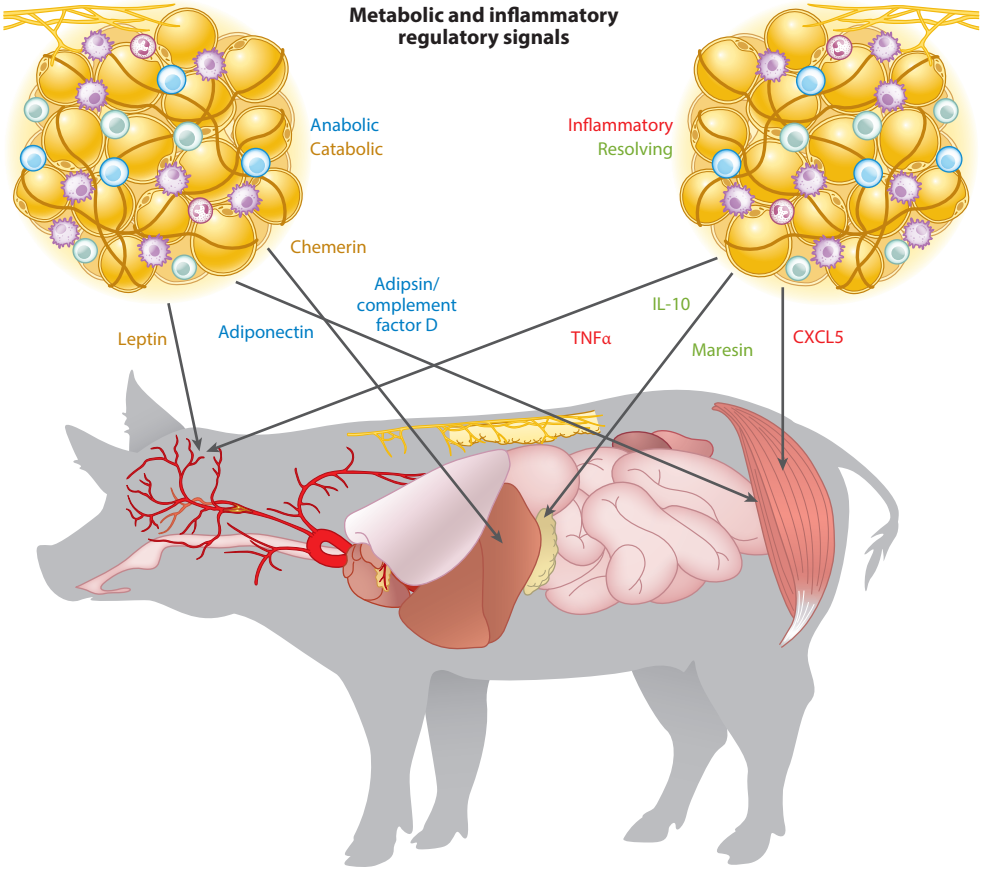

Figure 1

Adipose tissue serves as an important endocrine organ. Subcutaneous or visceral adipose tissue depots are found adjacent to most organs, with growing evidence of regional crosstalk. Additionally, endocrine factors with blended metabolic and immune control roles are released systemically, influencing all major organ systems. These factors use traditional endocrine control mechanisms (e.g., leptin) or provide suprahomeostatic control through inflammatory pathways (e.g., TNFα). Figure redrawn from images originally created in BioRender.

It is also worth noting that inflammatory signals are under the control of negative feedback loops. This is a core theme of classical endocrinology, where sensors detect deviations from homeostatic set points and hormone release is either stimulated or inhibited (i.e., negative feedback) depending on the direction of the deviation. In some more complex endocrine circuits like the somatotropic axis, these feedback loops include hormone inhibition of upstream-releasing hormones to keep the regulatory system in check. Most texts on immunity present inflammation in contrast to this, focusing on the feed-forward aspect of the cytokine storm; this is perhaps because the vast majority of research on inflammation has been in the context of pathological scenarios where normal negative feedback loops are failing. Therefore, there is a regulatory process involved in inflammatory circuits, although the precise mechanisms are poorly understood and likely differ from negative feedback loops described for classic endocrine circuits (50). However, as immunologists recognize, inflammation circuits follow an initial feed-forward signal amplification response

with a preprogrammed resolution phase. In fact, even isolated innate immune cells without the benefit of crosstalk with regulatory cells enter a resolution phase within hours after stimulation. Therefore, there is clearly a regulatory process involved in inflammatory circuits, and AT may play an important role in this regulatory system (51).

ADIPOSE TISSUE INFLAMMATION

As described in detail above, immune cells make up a significant portion of the cells in AT regardless of the relative health of an animal. However, the abundance, composition, and phenotypes of immune cells in AT vary greatly (52). A hallmark of inflamed AT is the increased population of ATMs and their shift from primarily M2 tissue repair phenotypes to more inflammatory M1 phenotypes (53). Crown-like structures are lipid-laden ATMs that appear around dying adipocytes in inflamed AT; they form this characteristic crown shape by surrounding and eventually engulfing much larger adipocytes, allowing for lysosomal degradation of the dead cell. These structures are associated with the appearance of multinucleated giant cells, which also contribute to clearance of dead adipocytes (54). The abundance of regulatory T cells declines (55) as dendritic cells become established (56) in the inflammatory AT environment, contributing to a shift from a balanced profile of pro- and anti-inflammatory immune cells to a population dominated by pro-inflammatory cells.

In association with an increasingly pro-inflammatory immune cell lineup, inflamed AT also features shifts in adipocyte function and signaling (57). These changes include enhanced lipolytic rate, reduced insulin sensitivity, changes in profiles of signaling lipids released (with paracrine and endocrine effects), and altered adipokine profiles. However, the factors that initiate the spiraling inflammatory adipocyte–immune cell crosstalk in metabolic disease (e.g., obesity) remain an enigma. One surprising finding is that, in contrast to the strong association between AT inflammation and insulin resistance, inducible suppression of inflammation specifically in adipocytes worsened systemic insulin resistance in obese mice (58). This may be due to the importance of inflammatory signaling for AT tissue expansion, including adipogenesis (59), which enables AT to continue removing nutrients from circulation for storage. In a very different AT scenario, lipodystrophy results in a marked systemic inflammation as huge numbers of macrophages are recruited around dying adipocytes (60), perhaps pointing to the importance of adipocyte death in driving AT inflammation.

Metabolic stress driven by chronic overnutrition is the best-documented situation that can promote AT inflammation. The adipose organ's inflammatory responses during these states are well characterized, contributing to the metabolic syndrome conditions that often accompany obesity, if not contributing to the obese state itself.

Systemic Consequences of Adipose Tissue Inflammation

In human metabolic syndrome and in many animal models developed for studying this condition, numerous risk factors and overlapping physiological consequences have made it difficult to pinpoint the initial domino that begins the etiological cascade toward diseases such as nonalcoholic fatty liver disease, diabetes, and hypertension. For example, feeding high-fat diets to mice with particular genetic backgrounds as a model for diet-induced obesity often simultaneously induces intestinal dysbiosis, multi-organ mitochondrial dysfunction, and AT inflammation. However, recent evidence—although not ruling out other initial triggers in different scenarios—suggests that AT inflammation induced by spontaneous hyperphagia is sufficient to impair liver health and systemic glucose homeostasis (61). In an ovariectomized mouse model of sarcopenia, inducible

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

deletion of HIF1α specifically in adipocytes was sufficient to reduce muscle inflammation (62). Further evidence for the key role of AT in systemic metabolism is the observation that small interfering RNA nanoparticles targeting knockdown of TNFα or osteopontin exclusively in epidydimal ATMs resulted in improved systemic glucose tolerance, at least in part through improve AT insulin sensitivity (63). Finally, interleukin (IL)-6 derived from brown AT was an essential mediator connecting psychological stress and enhanced inflammatory organ damage following lipopolysaccharide (LPS) challenge (64). Clearly, an inflammatory AT state can have marked effects on other organs and systemic physiology.

ADIPOSE TISSUE INFLAMMATION AND DISEASE RISK

A disease state—whether acute or chronic—is characterized by a loss of homeostasis. The inflammatory AT phenotype described above certainly limits the metabolic flexibility of an animal, at a minimum by constraining the buffering capability of AT and thereby allowing greater systemic perturbations in metabolite concentrations. This decrease in buffering capacity available for use by endocrine control circuits may well contribute to an ongoing reliance on inflammatory mechanisms to drive regulated variables back toward set points (46), leading to a worsening of the situation over time. For example, AT insulin resistance can disrupt glucose homeostasis, especially in monogastrics. In the short term, this leads to an adaptive inflammatory response (largely IL-1β-mediated) in pancreatic islets that promotes beta cell hyperplasia and hyperinsulinemia to overcome insulin resistance. However, this process eventually leads to beta cell dysfunction and death (65).

The altered AT state, with a bias toward pro-inflammatory resident immune cells and phenotypes, may result in hyperinflammatory responses to tissue damage or pathogen exposure. Conversely, some infectious diseases may lay the groundwork for an inflammatory AT phenotype, simply by promoting expansion of innate leukocyte populations, particularly in the context of a lipolytic response accompanying hypophagia. We consider both of these potential connections between disease and AT biology in the context of common health problems in livestock.

Adipose Tissue Responses to Metabolic Stress

As an essential organ for buffering the ebbs and flows of nutrient availability across months and years, AT is directly involved in adaptations to metabolic stress. However, new insights into signaling roles of AT have further highlighted its importance in determining whether animals are resilient or fragile in the face of such stressors.

Overnutrition. A general theme in obesity research has been the substantial increase in ATM populations in obese AT, along with a shift to more pro-inflammatory M1 phenotypes in these populations, which may contribute to metabolic disease (53). Although diseases associated with overnutrition are not prevalent in production animal species, specific examples demonstrate a close interaction between AT inflammation and the development of cardiometabolic diseases. Beef steers are often fed energy-dense rations during the finishing stages of feeding and are susceptible to the development of pulmonary hypertension and congestive heart failure, including right-sided congestive heart failure (66). Although the most recent data show that mortality risk from this condition is low [8 cases/10,000 cattle (67)], it appears to have increased from 2/10,000 in 2004, and its morbidity is unknown (66). Krafsur et al. (68) provide a detailed characterization of the histopathological findings in steers with hypertension and heart disease. Remarkably, the vascular remodeling of pulmonary vessels and especially the inflammatory responses observed in the perivascular AT resemble those observed in humans with the same disease. The mechanisms

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

that lead steers in the finishing stages of their production cycle to be more susceptible to this cardiovascular disease are unclear, but the perivascular and pericardial AT depots are part of the pathogenesis, and their roles warrant further research.

Reduction in dietary energy supply. Work in obese mice makes it tempting to draw a direct line between ATMs and metabolic disease. However, a recent experiment in mice clearly dissociated total ATM abundance from metabolic dysfunction (69). In this study, mice were exposed to high-fat (60% of calories) diets to induce obesity and accumulated a predictable increase in ATMs. However, a subset of these mice were then reverted back to normal chow, resulting in a marked decrease in body weight, reductions in adipose and liver mass, and recovery of glucose tolerance (69). Unexpectedly, the population of ATMs increased further, rather than decreasing as might be expected in correlation with improved metabolic health. Furthermore, mice undergoing diet-induced weight loss showed an increase in macrophage-mediated $CD8^+$ T cell inflammation for at least a month after the diet change (69).

In fact, during extended periods of negative energy balance (e.g., prolonged fasting), intense lipid mobilization characterized by limited lipogenesis and high rates of lipolysis triggers an inflammatory response within the AT. This response is observed in humans (70), rodents (71), and dairy cows (72) and is characterized by enhanced trafficking of macrophages and other immune cells, often resulting in a doubling of ATM populations in both internal and subcutaneous depots during the period of peak lipolysis. Several mechanisms trigger inflammation during lipolysis. (*a*) Lipolysis products are potent mediators of inflammation. As shown in rodents (73) and cows (74), the breakdown of triglycerides by adipocytes' adipose triglyceride lipase and hormone-sensitive lipase releases arachidonic and linoleic fatty acids that are substrates for oxylipin biosynthesis. Oxylipins include prostaglandins, thromboxanes, leukotrienes, and oxidized linoleic acid metabolites that are potent chemoattractants and initiators of inflammatory processes. (*b*) Lipolysis products such as neutral lipids are toxic to the cells. For example, ceramide-induced lipotoxicity affects mitochondrial function by impairing oxidative phosphorylation and promoting mitochondrial fragmentation [extensively reviewed by Chaurasia & Summers (75) and with emphasis on dairy cattle research by McFadden & Rico (76)]. (*c*) Adipocyte pyroptosis, an inflammation-induced programmed cell death, is observed during extended periods of lipolysis, releasing lipid droplet contents that trigger inflammation and lipotoxicity (77). How permanent is AT inflammation after the period of negative energy balance? This has not been studied thoroughly across species, but ATMs declined as obese, energy-restricted mice returned to a lean body weight, resulting in similar ATM populations as in control (chow diet) mice after 6 to 8 weeks of caloric restriction (71).

These findings point to the subtleties of inflammation–metabolism links. Although chronic overnutrition can clearly promote a pro-inflammatory environment and immune cell accumulation within AT, immune cells are also important mediators of AT lipid mobilization. For example, ATMs secrete potent lipolytic agents such as IL-6 and catecholamines (78, 79). How are we to make sense of the ATM accumulation during the long-term anabolic state that promotes obesity and also the transient increase in ATMs during the catabolic state induced by caloric restriction?

Perhaps there is a consistent connection to lipolytic rate, despite the apparent differences in net AT accretion in these two scenarios. Although not carefully studied in domestic animals, human obesity clearly causes a sustained increase in plasma FFA concentration as well as increased FFA turnover rate in abdominal obesity (80). Additionally, obese mouse AT releases triglyceride-loaded exosomes at nearly double the rate of lean mouse AT (81). This suggests an upregulation of futile cycling in obese AT, with high rates of lipid export from adipocytes coupled with reesterification,

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

though still resulting in sustained increases in lipolytic end products and exosomes with signaling effects. In this paradigm, ATMs may help prevent AT lipotoxicity during periods of rapid weight loss by cleaning up both excess lipids (82) and atrophied, dying adipocytes, as well as by modulating the rate of lipolysis to avoid systemic metabolic risks. However, in sustained overnutrition, these regulatory roles may be overwhelmed by the sheer increase in AT immune cell populations, leading to pro-inflammatory crosstalk locally and eventually systemic effects.

Onset of lactation. Lactation is one of the most energy-demanding processes that a mammal can undergo, and this is made even more extreme by genetic selection for high milk production in dairy species. In dairy cows, around parturition and during the initial stages of lactation, AT provides fatty acids to counteract the reduced energy availability caused by lactogenesis and appetite suppression. This lactation-associated negative energy balance triggers an increase in lipolysis, which involves the activation of adipocyte lipases through hormonal signaling (e.g., growth hormone) and neuronal stimulation (e.g., catecholamines). During the first 40 days after calving, lipolysis leads to a 25–35% reduction of the AT mass (83), leading to a remodeling process that includes inflammatory responses and immune cell infiltration, with ATMs playing a crucial role in this process in dairy cows (84). Although not demonstrated in sheep and goats, females from these species likely exhibit similar remodeling processes in their AT postpartum.

The induction of immune cell activity in response to both chronic overnutrition and rapid tissue mobilization raises the possibility that weight cycling may promote even more dramatic AT inflammation. In an experiment with mice, Anderson et al. (85) demonstrated that, compared to mice fed to a similar body composition on a delayed high-fat diet treatment, mice who were made obese, then calorie restricted, and finally fed to cycle back to obesity showed a clear increase in $CD4^+$ and $CD8^+$ T cells in AT. This accumulation of T cells drove increased AT expression of Th1 cytokines like *IFNG* and was associated with reduced AT insulin sensitivity and systemic glucose tolerance. Subsequent work confirmed these findings and also demonstrated mast cell accumulation in response to weight cycling (86). The alteration of AT function in response to weight cycling subsequently promotes fatty liver disease (87), and corroborating epidemiological evidence suggests that high body weight variability (upper quartile) in humans is associated with increased all-cause mortality (88).

Weight cycling is common in both wild and domestic species, due to seasonal variation in temperature and nutrient availability and/or to variation in energy demands across production cycles. Dairy cattle provide an excellent example of this cyclicity; most cows selected for high levels of milk production mobilize large quantities of stored AT lipids as well as skeletal muscle in the first month of lactation (in some cases >10% of body weight is lost), followed by repletion of body stores in late lactation. However, the magnitude of these shifts varies widely among cows. Zachut & Moallem (89) reported that some cows consistently lose more body weight in the first month of lactation than others, and the magnitude of loss is amplified with increasing parity. The high-weight-loss cows showed clear signs of glucose intolerance as well as inferior fertility. Furthermore, large data sets have demonstrated that greater body condition at parturition is associated with greater loss of body condition postpartum (90), indicative of more dramatic weight cycling; this dynamic was also found to have a strong genetic component.

These findings point to a subset of dairy cows with progressively more dramatic weight cycling as they progress from first parity through subsequent lactations. Do these cows continually accrue additional AT immune cells with increasingly pro-inflammatory phenotypes in this process? Such a phenomenon may help to explain the greater hazard of not only metabolic diseases (91) but also infectious diseases (92, 93) and infertility (94) in cows that experience greater loss of body condition in early lactation (**Figure 2**).

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

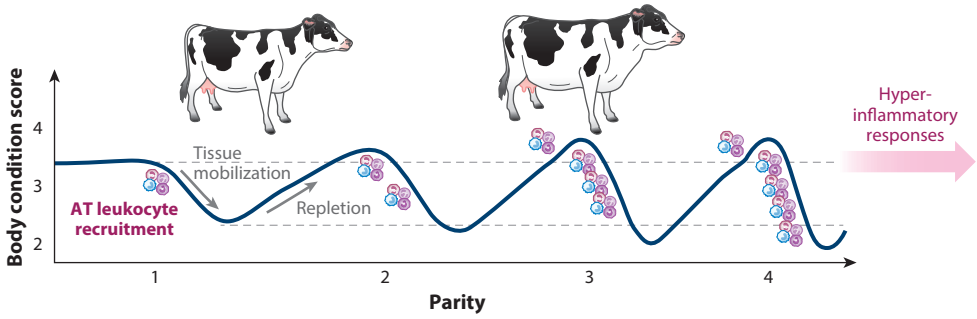

Figure 2

A proposed role for weight cycling–induced adipose tissue (AT) immune activation in periparturient dairy cow diseases. Subclinical or clinical metabolic and/or infectious diseases affect more than 40% of dairy cows in the month after parturition, and the risk of most such diseases increases progressively with increasing parity. The weeks after parturition are a time of rapid tissue mobilization due to nutrient demands for lactation outstripping supply. During this rapid adipose tissue lipolysis, recruitment of macrophages and other leukocytes makes adipose more immune reactive. Evidence from mouse models suggests that some, but not all, of this accumulation of immune cells is depleted over the ~9-month repletion phase. In particular, $CD4^+$ and $CD8^+$ T cells may be retained until the subsequent parturition, leading to an exacerbated immune cell recruitment during the second period of rapid lipolysis. In some cows, subsequent lactations show more extreme swings, with both greater peak and nadir body condition scores. By parity 4, immune cell recruitment may have reached a magnitude where metabolic and immunological impacts of the inflamed AT reduce the cow's resilience to disease challenges. Figure redrawn from images originally created in BioRender.

Mechanistically, links between body condition loss and disease could be mediated by impacts of lipolytic products and downstream metabolites on innate and adaptive immune cell function (95), resulting in inferior pathogen clearance (96); the concentrations of these metabolites are increased by an inflammatory AT environment (**Figure 3**). An inflammatory AT environment could also have direct negative effects on adaptive (97) and innate immune function.

A study comparing AT transcriptomes across divergent sheep breeds in summer and winter suggested that fat-tailed sheep breeds—known for resilience in extreme environments—had less lipolysis, AT remodeling, and immune cell infiltration during the nutrient-sparse winter months (98). These findings suggested a greater capacity of the fat-tailed sheep to adapt to the harsh conditions using endocrine regulatory processes rather than inflammatory mechanisms, and they align with the concept that lesser AT cyclicity may support overall resilience of the animal in the face of stressors.

The impact of weight cycling on AT immune crosstalk may be an underlying factor explaining why, at least in humans, AT appears to show more dramatic aging-induced increases in inflammatory tone than muscle (99). To our knowledge, aging-associated inflammation has not been examined carefully in food animals, but increasing AT inflammation may be a causal factor underlying the greater risk of postpartum disease in dairy cattle of increasing parities (100).

Ketosis and pregnancy toxemia. In cows, does, and ewes, pre- or postpartum energy deficits can lead to a metabolic state characterized by hypoglycemia, hyperketonemia, and elevated FFAs (101, 102). In cows, this state is termed clinical ketosis and has greatest prevalence during the first two weeks after calving. In does and ewes, the metabolic disease is known as pregnancy toxemia and occurs most commonly in the last three weeks of gestation. In these diseases, excessive and protracted AT lipolysis is a determinant factor. Bovine clinical ketosis is associated with AT

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

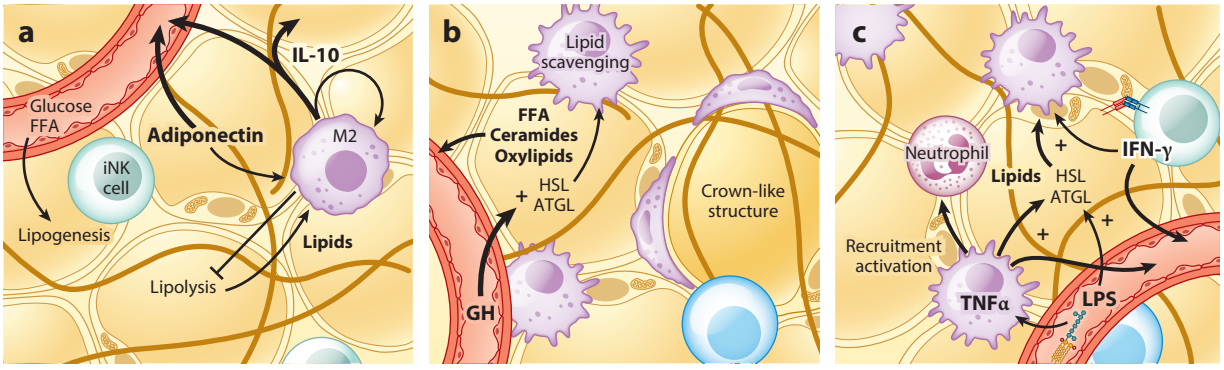

Figure 3

Cellular composition and physiological stressors influence adipose tissue signaling. Hormones and metabolites released by both adipocytes and neighboring cells have autocrine, paracrine, and endocrine roles, and metabolic and disease stressors have overlapping effects. (*a*) Healthy adipose tissue contains a modest abundance of various types of immune cells, primarily with regulatory and debris-scavenging phenotypes. Macrophages are primarily of the alternative activation phenotype (M2), producing anti-inflammatory IL-10, as can iNK cells. Macrophages also take up free lipids released by steady-state lipolysis and provide negative feedback on that pathway. Adiponectin, produced by adipocytes, promotes lipogenesis and has anti-inflammatory effects locally and systemically. (*b*) In response to nutrient deficiency, systemic endocrine signals like GH activate key lipolytic enzymes, releasing lipids that serve as both fuels and signals for local and distant cells. High rates of lipolytic product exposure can cause phenotype switching of macrophages (M1), and inflammatory signaling by local T cells helps recruit additional macrophages to the tissue. As some adipocytes become depleted, the rate of cell death increases, and macrophages and other cell types contribute to removal of the cell and debris to limit release of damage molecules. (*c*) In a disease scenario, microbial-associated molecules like LPS result in resident immune cell activation and additional recruitment of leukocytes, potentially including neutrophils. LPS also acts to promote lipolysis, as do inflammatory cytokines (e.g., TNFα) released from immune cells. Additionally, iNK cells can be activated in part through sensing major histocompatibility complex II on adipocytes, triggering release of IFN-γ, which can prime macrophage responses and act systemically. Abbreviations: ATGL, adipose triglyceride lipase; FFA, free fatty acid; GH, growth hormone; HSL, hormone-sensitive lipase; IFN-γ, interferon-γ; iNK, invariant natural killer; LPS, lipopolysaccharide. Figure redrawn with images originally created in BioRender.

inflammation that encompasses ATM infiltration with M1 polarization and formation of crown-like structures (32). The AT of ketotic cows shows increased activity of signaling pathways involving pro-inflammatory cytokines and chemokines, such as NF-κB, MAPK, and Jak-STAT (103). These pathways sustain the inflammatory response by promoting immune cell movement into the AT.

Dry-off. The phase when dairy animals are not producing milk is referred to as the dry period. This phase of the lactation cycle begins at dry-off, which involves an abrupt cessation of milking that is followed by acute remodeling of the mammary gland (104). This remodeling is accompanied by hormonal, metabolic, and behavioral changes that induce short periods of limited energy supply. After abrupt cessation of milking, there is an increase in lipolysis in dairy cows, as evidenced by higher levels of FFA and β-hydroxybutyrate in the first 48 h post dry-off (105). Although there have been no reports on AT inflammation at dry-off or during the dry period, the short duration of intense lipolysis and the minimal changes in systemic inflammation biomarkers such as haptoglobin and albumin (105) suggest that inflammatory changes in ATs are unlikely in the initial days of the dry period.

Weaning. Weaning in mammalian species is a multifaceted process that encompasses the cessation of suckling and triggers various physiological, nutritional, and behavioral transformations in both the mother and offspring (106). The definition of weaning differs among production animal species. For instance, in dairy cows, the calves are separated from their mothers shortly after

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

birth, and weaning is typically defined as the removal of dietary milk or milk replacer. In meat-producing species, weaning involves not only the withdrawal of milk but also the separation of the mother from her offspring. Irrespective of the species, these biological and behavioral changes induce metabolic stress, and AT is a crucial component of the systemic responses that enable the weaned animal to adapt to its new metabolic status (25).

Shipping. Almost all production animals require transportation at a certain point in their productive cycle. Cattle require shipping by road more often than other species given the segmented nature of the dairy and beef industries in many countries (107). Transportation journeys involve feed restriction, psychological stress, and physical exertion that invariably will induce a metabolic response in AT (107). Immediately after transportation and up to 72 h post arrival, beef steers exhibited increased lipolysis rates as reflected in blood concentration of FFA, which only returned to pretransport levels 3 weeks later (108). Although there are no reports characterizing AT's inflammatory response during and after transportation, sustained lipolytic responses induced by feed restriction for more than 72 h induce macrophage infiltration into AT, as shown in lactating dairy cattle (72). Furthermore, high circulating FFA concentrations post transportation invariably coincide with increased blood concentrations of acute-phase proteins, including haptoglobin and serum amyloid A in beef steers (109, 110), that are positively associated with AT inflammation (111).

Heat stress. In food animal species, production efficiency increases when animals are kept at thermoneutrality. This metabolic state is the environmental temperature at which the animal does not have to expend energy to modulate core body temperature (112). When ambient temperature increases or decreases, thermal stress develops, inducing metabolic and inflammatory changes in the AT (113). The incidence of acute (hours) and chronic (days) heat stress in livestock farms is increasing due to rising global ambient temperature and increased frequency of severe heat waves (114). In pigs, AT's heat stress adaptation begins with a short-term rise (hours) in lipolysis that transitions into a long-term increase in triglyceride biosynthesis and lipogenesis (115–117). These changes in lipid mobilization patterns are coupled with a rise in leptin secretion, which in turn may enhance the feed intake depression observed during heat stress (116). In both beef and dairy cattle, heat stress induces an inflammatory response in AT that is characterized by increased expression of interleukins and interferon and acute-phase inflammation signaling (118, 119).

Role of Adipose Tissue in Responses to Infectious Disease

During bacterial infections, the AT is an integral part of the inflammatory response. Endotoxins such as LPS trigger an inflammatory reaction in adipocytes and neighboring immune cells (120). In dairy cows, endotoxin induces lipolysis in adipocytes in vitro (121); however, the lipolytic response in vivo is dynamic and includes a refractory stage during which fatty acid release from AT is reduced, coinciding with a peak in circulating ketone bodies (122, 123). As in negative-energy-balance states, LPS-induced lipolysis also activates an inflammatory response within AT that is characterized by immune cell infiltration. Although research on AT inflammatory and immune responses during infectious diseases is limited in livestock species, experiments in humans and rodents provide important insight into these responses. For example, one day after sepsis induction with intraperitoneal injection of fecal slurry in mice, the AT exhibits a massive infiltration of neutrophils, whereas populations of other immune cells are reduced (124). One month post intraperitoneal challenge, these mice showed increased migration of macrophages and T and NK cells, whereas neutrophil populations dwindled, though they remained higher than before sepsis induction (124). These responses all contribute to sterilization of AT during infection, and in fact

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

adipocytes may even contribute directly to pathogen clearance through production of nitric oxide, a free radical with bactericidal properties (125, 126). The AT is therefore an integral part of the response to bacterial infections by providing fatty acids as energy substrates, harboring immune cells, and generating bactericidal compounds.

Similar to bacterial infections, during viral diseases, the AT is a viral target and a site of host immune and inflammatory responses. In human and simian immunodeficiency infections, both viruses attack adipocytes, macrophages, and $CD4^+$ T cells in AT, as shown by viral DNA and RNA detection (127), and a similar process occurs in lymphocytic choriomeningitis virus infection of mice (128). These AT cells become a reservoir for these viruses and sustain chronic inflammatory responses during infection. The AT is also a target of parasitic diseases, and the most well-studied is trypanosomiasis (i.e., Chagas disease). During infection, the trypanosoma invades adipocytes and persists in the cells for long periods of time (129). Parasites living in AT can infect naïve animals (130).

Because AT depots are located throughout the body, surrounding major organs such as the kidneys and heart, they are major integrators of inflammatory signals during infectious diseases. Integration mechanisms include the secretion of adipokines, inflammatory cytokines, fatty acids, and lipid mediators of inflammation. For instance, in individuals with critical illness (i.e., sepsis), the blood concentrations of adipokines such as leptin, resistin, and visfatin surge during the initial days of the illness, whereas adiponectin levels decrease (131). This particular profile of adipokines inhibits insulin sensitivity in adipocytes, promoting energy allocation for immune cell function (132). Adipocytes also secrete IL-6 during acute inflammation, further prioritizing energy for immune responses (133).

Because many of the mechanistic links between AT physiology and immune function are recent discoveries, few livestock studies have explored whether changes in immune or inflammatory programming of AT predispose animals to subsequent disease or alter productivity. In one intriguing study, growing pigs from lines selected for high or low feed efficiency were exposed to good or poor hygiene conditions. These factors interacted: Pigs with low-feed-efficiency genetics exposed to poor hygiene had the poorest growth rate and also the greatest markers of AT oxidative stress, which was not the case in liver tissue (134). Although very preliminary, such findings suggest that early-life pathogen pressure may interact with genetic background to influence AT microenvironment and subsequently growth performance.

CONCLUSIONS

In livestock species, AT has recognized importance due to its impact on carcass value in meat animals and as a nutrient reservoir for times of nutrient imbalance. However, its roles in regulating whole-body physiological functions like nutrient metabolism and immunity are still being described. In this review, we present preliminary evidence that—due to its regulatory role in these physiological systems—AT can be a key anatomic site where life history, including adiposity and disease events, impacts subsequent responses to stressors. Inflamed AT is a consequence of an initial insult or stressor, but it can result in a fragile physiological state that makes animals susceptible to subsequent challenges. In this way, the immunometabolic wiring of AT is both a consequence of and a precursor to disease.

Promoting a healthy AT environment with a pro-resolution immune cell roster rather than an inflammatory environment may not only help to prevent metabolic disorders but also enable more appropriate responses to infectious agents. We propose several key questions that should be addressed in domestic species to shed more light on the potential utility of AT-focused management strategies to promote long-term health.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

FUTURE ISSUES

1. Do infections early in life affect the cellular composition of SCAT or VAT depots?
2. To what extent do AT immune cell populations and phenotypes influence responses to metabolic, environmental, and pathogenic stressors that commonly affect food animal species?
3. Can dietary or pharmacological treatments during or after stressors reprogram AT physiology to improve response to subsequent challenges?
4. Can we influence AT biology in a way that enhances immune memory, including the potential to enhance vaccine efficacy?
5. Do livestock breeds or strains with less cyclical change in body fat (across production cycles or seasons) maintain more balanced AT and systemic inflammatory tone and thereby retain greater resilience to stressors as they age?

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

LITERATURE CITED

1. Zhu Q, Scherer PE. 2018. Immunologic and endocrine functions of adipose tissue: implications for kidney disease. *Nat. Rev. Nephrol.* 14(2):105–20
2. Weber JM. 2011. Metabolic fuels: regulating fluxes to select mix. *J. Exp. Biol.* 214(Pt. 2):286–94
3. Froy O, Garaulet M. 2018. The circadian clock in white and brown adipose tissue: mechanistic, endocrine, and clinical aspects. *Endocr. Rev.* 39(3):261–73
4. Börgeson E, Boucher J, Hagberg CE. 2022. Of mice and men: pinpointing species differences in adipose tissue biology. *Front. Cell Dev. Biol.* 10:1003118
5. Hammoud SH, AlZaim I, Al-Dhaheri Y, Eid AH, El-Yazbi AF. 2021. Perirenal adipose tissue inflammation: novel insights linking metabolic dysfunction to renal diseases. *Front. Endocrinol.* 12:707126
6. Moritz E, Dadson P, Saukko E, Honka M-J, Koskensalo K, et al. 2022. Renal sinus fat is expanded in patients with obesity and/or hypertension and reduced by bariatric surgery associated with hypertension remission. *Metabolites* 12(7):617
7. Ahmad MF, Ferland D, Ayala-Lopez N, Contreras GA, Darios E, et al. 2019. Perivascular adipocytes store norepinephrine by vesicular transport. *Arterioscler. Thromb. Vasc. Biol.* 39(2):188–99
8. Yang L, Chen L, Lu X, Tan A, Chen Y, et al. 2018. Peri-ovarian adipose tissue contributes to intraovarian control during folliculogenesis in mice. *Reproduction* 156(2):133–44
9. Bornelöv S, Seroussi E, Yosefi S, Benjamini S, Miyara S, et al. 2018. Comparative omics and feeding manipulations in chicken indicate a shift of the endocrine role of visceral fat towards reproduction. *BMC Genom.* 19:295
10. Billon N, Iannarelli P, Monteiro MC, Glavieux-Pardanaud C, Richardson WD, et al. 2007. The generation of adipocytes by the neural crest. *Development* 134(12):2283–92
11. Louveau I, Perruchot M-H, Bonnet M, Gondret F. 2016. Invited review: pre- and postnatal adipose tissue development in farm animals: from stem cells to adipocyte physiology. *Animal* 10(11):1839–47
12. Hausman GJ, Kauffman RG. 1986. The histology of developing porcine adipose tissue. *J. Anim. Sci.* 63(2):642–58
13. Wright JT, Hausman GJ. 1990. Adipose tissue development in the fetal pig examined using monoclonal antibodies. *J. Anim. Sci.* 68(4):1170–75

14. Hausman GJ, Thomas GB. 1984. The development of the inner layer of backfat in fetal and young pigs. *J. Anim. Sci.* 58(6):1550–60
15. Casteilla L, Forest C, Robelin J, Ricquier D, Lombet A, Ailhaud G. 1987. Characterization of mitochondrial-uncoupling protein in bovine fetus and newborn calf. *Am. J. Physiol.* 252(5 Pt. 1):E627–36
16. Bonnet M, Cassar-Malek I, Chilliard Y, Picard B. 2010. Ontogenesis of muscle and adipose tissues and their interactions in ruminants and other species. *Animal* 4(7):1093–109
17. Gemmell RT, Alexander G. 1978. Ultrastructural development of adipose tissue in foetal sheep. *Aust. J. Biol. Sci.* 31(5):505–15
18. Robelin J, Chilliard Y. 1989. Short-term and long-term effects of early nutritional deprivation on adipose tissue growth and metabolism in calves. *J. Dairy Sci.* 72(2):505–13
19. Wijayasinghe MS, Smith NE, Baldwin RL. 1986. Effects of age, milk replacer composition, and rumen function on lipogenesis in adipose tissue of young calves. *J. Dairy Sci.* 69(9):2358–69
20. Bispham J, Gopalakrishnan GS, Dandrea J, Wilson V, Budge H, et al. 2003. Maternal endocrine adaptation throughout pregnancy to nutritional manipulation: consequences for maternal plasma leptin and cortisol and the programming of fetal adipose tissue development. *Endocrinology* 144(8):3575–85
21. Edwards LJ, McFarlane JR, Kauter KG, McMillen IC. 2005. Impact of periconceptional nutrition on maternal and fetal leptin and fetal adiposity in singleton and twin pregnancies. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* 288(1):R39–45
22. Wagoner B, Hausman DB, Harris RBS. 2006. Direct and indirect effects of leptin on preadipocyte proliferation and differentiation. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* 290(6):R1557–64
23. Desai M, Ross MG. 2011. Fetal programming of adipose tissue: effects of intrauterine growth restriction and maternal obesity/high-fat diet. *Semin. Reprod. Med.* 29(3):237–45
24. Leachman JR, Cincinelli C, Ahmed N, Dalmasso C, Xu M, et al. 2022. Early life stress exacerbates obesity in adult female mice via mineralocorticoid receptor-dependent increases in adipocyte triglyceride and glycerol content. *Life Sci*. 304:120718
25. Li Y, Thelen KM, Fernández KM, Nelli R, Fardisi M, et al. 2022. Developmental alterations of intestinal SGLT1 and GLUT2 induced by early weaning coincides with persistent low-grade metabolic inflammation in female pigs. *Am. J. Physiol. Liver Physiol.* 322(3):G346–59
26. Michelotti TC, Kisby BR, Flores LS, Tegeler AP, Fokar M, et al. 2022. Single-nuclei analysis reveals depot-specific transcriptional heterogeneity and depot-specific cell types in adipose tissue of dairy cows. *Front. Cell Dev. Biol.* 10:1025240
27. Emont MP, Jacobs C, Essene AL, Pant D, Tenen D, et al. 2022. A single-cell atlas of human and mouse white adipose tissue. *Nature* 603(7903):926–33
28. Rigamonti A, Brennand K, Lau F, Cowan CA. 2011. Rapid cellular turnover in adipose tissue. *PLOS ONE* 6(3):e17637
29. Spalding KL, Arner E, Westermark PO, Bernard S, Buchholz BA, et al. 2008. Dynamics of fat cell turnover in humans. *Nature* 453(7196):783–87
30. Jin L, Tang Q, Hu S, Chen Z, Zhou X, et al. 2021. A pig BodyMap transcriptome reveals diverse tissue physiologies and evolutionary dynamics of transcription. *Nat. Commun.* 12:3715
31. Aylward BA, Clark ML, Galileo DS, Baernard AM, Wilson JR, et al. 2019. Immune cell populations residing in mesenteric adipose depots and mesenteric lymph nodes of lean dairy cows. *J. Dairy Sci.* 102(4):3452–68
32. Contreras GA, Kabara E, Brester J, Neuder L, Kiupel M. 2015. Macrophage infiltration in the omental and subcutaneous adipose tissues of dairy cows with displaced abomasum. *J. Dairy Sci.* 98(9):6176–87
33. Russo L, Lumeng CN. 2018. Properties and functions of adipose tissue macrophages in obesity. *Immunology* 155(4):407–17
34. Kratz M, Coats BR, Hisert KB, Hagman D, Mutskov V, et al. 2014. Metabolic dysfunction drives a mechanistically distinct proinflammatory phenotype in adipose tissue macrophages. *Cell Metab*. 20(4):614–25
35. Harasymowicz NS, Rashidi N, Savadipour A, Wu C-L, Tang R, et al. 2021. Single-cell RNA sequencing reveals the induction of novel myeloid and myeloid-associated cell populations in visceral fat with long-term obesity. *FASEB J.* 35(3):e21417

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

36. Oliveira BM, Pinto A, Correia A, Ferreira PG, Vilanova M, Teixeira L. 2020. Characterization of myeloid cellular populations in mesenteric and subcutaneous adipose tissue of Holstein-Friesian cows. *Sci. Rep.* 10:1771
37. Oliveira BM, Rasteiro AM, Correia A, Pinto A, Meireles P, et al. 2019. T cells in mesenteric and subcutaneous adipose tissue of Holstein-Friesian cows. *Sci. Rep.* 9:3413
38. Han S, Zaretsky AG, Andrade-Oliveira V, Brenchley JM, Brodsky IE, et al. 2017. The white adipose tissue is a reservoir for memory T cells and promotes protective memory responses to infection. *Immunity* 47(6):1154–68.e6
39. Torchon E, Ray R, Hulver MW, McMillan RP, Voy BH. 2017. Fasting rapidly increases fatty acid oxidation in white adipose tissue of young broiler chickens. *Adipocyte* 6(1):33–39
40. Czech MP. 2020. Mechanisms of insulin resistance related to white, beige, and brown adipocytes. *Mol. Metab.* 34:27–42
41. Renner S, Blutke A, Dobenecker B, Dhom G, Müller TD, et al. 2018. Metabolic syndrome and extensive adipose tissue inflammation in morbidly obese Göttingen minipigs. *Mol. Metab.* 16:180–90
42. Kennedy GC. 1953. The role of depot fat in the hypothalamic control of food intake in the rat. *Proc. R. Soc. Lond. B* 140(901):578–96
43. Friedman JM, Halaas JL. 1998. Leptin and the regulation of body weight in mammals. *Nature* 395(6704):763–70
44. Gómez-Banoy N, Guseh JS, Li G, Rubio-Navarro A, Chen T, et al. 2019. Adipsin preserves beta cells in diabetic mice and associates with protection from type 2 diabetes in humans. *Nat. Med.* 25(11):1739–47
45. Shim K, Begum R, Yang C, Wang H. 2020. Complement activation in obesity, insulin resistance, and type 2 diabetes mellitus. *World J. Diabetes* 11(1):1–12
46. **Meizlish ML, Franklin RA, Zhou X, Medzhitov R. 2021. Tissue homeostasis and inflammation. *Annu. Rev. Immunol.* 39:557–81**
47. Wang A, Huen SC, Luan HH, Yu S, Zhang C, et al. 2016. Opposing effects of fasting metabolism on tissue tolerance in bacterial and viral inflammation. *Cell* 166(6):1512–25.e12
48. Cruz-Pineda WD, Parra-Rojas I, Rodríguez-Ruíz HA, Illades-Aguiar B, Matia-García I, Garibay-Cerdenares OL. 2022. The regulatory role of insulin in energy metabolism and leukocyte functions. *J. Leukoc. Biol.* 111(1):197–208
49. Šestan M, Marinović S, Kavazović I, Cekinović Đ, Wueest S, et al. 2018. Virus-induced interferon-γ causes insulin resistance in skeletal muscle and derails glycemic control in obesity. *Immunity* 49(1):164–77.e6
50. Tisoncik JR, Korth MJ, Simmons CP, Farrar J, Martin TR, Katze MG. 2012. Into the eye of the cytokine storm. *Microbiol. Mol. Biol. Rev.* 76(1):16–32
51. Bradford BJ, Swartz TH. 2020. Review: following the smoke signals: inflammatory signaling in metabolic homeostasis and homeorhesis in dairy cattle. *Animal* 14(S1):s144–54
52. **Trim WV, Lynch L. 2022. Immune and non-immune functions of adipose tissue leukocytes. *Nat. Rev. Immunol.* 22(6):371–86**
53. Olefsky JM, Glass CK. 2010. Macrophages, inflammation, and insulin resistance. *Annu. Rev. Physiol.* 72:219–46
54. Braune J, Lindhorst A, Fröba J, Hobusch C, Kovacs P, et al. 2021. Multinucleated giant cells in adipose tissue are specialized in adipocyte degradation. *Diabetes* 70:538–48
55. Feuerer M, Herrero L, Cipolletta D, Naaz A, Wong J, et al. 2009. Lean, but not obese, fat is enriched for a unique population of regulatory T cells that affect metabolic parameters. *Nat. Med.* 15(8):930–39
56. Li C, Wang G, Sivasami P, Ramirez RN, Zhang Y, et al. 2021. Interferon-α-producing plasmacytoid dendritic cells drive the loss of adipose tissue regulatory T cells during obesity. *Cell Metab.* 33(8):1610–23.e5
57. Blaszczak AM, Jalilvand A, Hsueh WA. 2021. Adipocytes, innate immunity and obesity: a mini-review. *Front. Immunol.* 12:1778
58. Zhu Q, An YA, Kim M, Zhang Z, Zhao S, et al. 2020. Suppressing adipocyte inflammation promotes insulin resistance in mice. *Mol. Metab.* 39:101010
59. Vishvanath L, Gupta RK. 2019. Contribution of adipogenesis to healthy adipose tissue expansion in obesity. *J. Clin. Investig.* 129(10):4022–31

46. Provides a conceptual framework to understand physiological regulation by inflammatory signaling pathways.

52. A comprehensive overview of current knowledge on adipose tissue immune cells and their roles.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

60. Herrero L, Shapiro H, Nayer A, Lee J, Shoelson SE. 2010. Inflammation and adipose tissue macrophages in lipodystrophic mice. *PNAS* 107(1):240–45
61. Petrick HL, Foley KP, Zlitni S, Brunetta HS, Paglialunga S, et al. 2020. Adipose tissue inflammation is directly linked to obesity-induced insulin resistance, while gut dysbiosis and mitochondrial dysfunction are not required. *Function* 1(2):zqaa013
62. Ge Y, Li S, Yao T, Tang Y, Wan Q, et al. 2023. Promotion of healthy adipose tissue remodeling ameliorates muscle inflammation in a mouse model of sarcopenic obesity. *Front. Nutr.* 10:1065617
63. Aouadi M, Tencerova M, Vangala P, Yawe JC, Nicoloro SM, et al. 2013. Gene silencing in adipose tissue macrophages regulates whole-body metabolism in obese mice. *PNAS* 110(20):8278–83
64. Qing H, Desrouleaux R, Israni-Winger K, Mineur YS, Fogelman N, et al. 2020. Origin and function of stress-induced IL-6 in murine models. *Cell* 182(2):372–87.e14
65. Guo J, Fu W. 2020. Immune regulation of islet homeostasis and adaptation. *J. Mol. Cell Biol.* 12(10):764–74
66. Neary JM, Booker CW, Wildman BK, Morley PS. 2016. Right-sided congestive heart failure in North American feedlot cattle. *J. Vet. Intern. Med.* 30(1):326–34
67. Johnson BT, White BJ, Dahlman DA, Rademacher RD, Amrine DE, Larson RL. 2023. Retrospective analysis of risk and timing of death from right heart failure in United States and Canadian feedlot necropsied cattle. *J. Am. Vet. Med. Assoc.* 261(9):1–7
68. Krafsur GM, Neary JM, Garry F, Holt T, Gould DH, et al. 2018. Cardiopulmonary remodeling in fattened beef cattle: a naturally occurring large animal model of obesity-associated pulmonary hypertension with left heart disease. *Pulm. Circ.* 9(1):2045894018796804
69. Surendar J, Karunakaran I, Frohberger SJ, Koschel M, Hoerauf A, Hübner MP. 2020. Macrophages mediate increased CD8 T cell inflammation during weight loss in formerly obese mice. *Front. Endocrinol.* 11:257
70. Fazeli PK, Zhang Y, O'Keefe J, Pesaresi T, Lun M, et al. 2020. Prolonged fasting drives a program of metabolic inflammation in human adipose tissue. *Mol. Metab.* 42:101082
71. Kosteli A, Sugaru E, Haemmerle G, Martin JF, Lei J, et al. 2010. Weight loss and lipolysis promote a dynamic immune response in murine adipose tissue. *J. Clin. Investig.* 120(10):3466–79
72. Contreras GA, Thelen K, Schmidt SE, Strieder-Barboza C, Preseault CL, et al. 2016. Adipose tissue remodeling in late-lactation dairy cows during feed-restriction-induced negative energy balance. *J. Dairy Sci.* 99(12):10009–21
73. Gartung A, Zhao J, Chen S, Mottillo E, VanHecke GC, et al. 2016. Characterization of eicosanoids produced by adipocyte lipolysis: implication of cyclooxygenase-2 in adipose inflammation. *J. Biol. Chem.* 291(31):16001–10
74. Contreras GA, De Koster J, de Souza J, Laguna J, Mavangira V, et al. 2020. Lipolysis modulates the biosynthesis of inflammatory lipid mediators derived from linoleic acid in adipose tissue of periparturient dairy cows. *J. Dairy Sci.* 103(2):1944–55
75. Chaurasia B, Summers SA. 2021. Ceramides in metabolism: key lipotoxic players. *Annu. Rev. Physiol.* 83:303–30
76. McFadden JW, Rico JE. 2019. Invited review: sphingolipid biology in the dairy cow: the emerging role of ceramide. *J. Dairy Sci.* 102(9):7619–39
77. Wang Y, Kanneganti T-D. 2021. From pyroptosis, apoptosis and necroptosis to PANoptosis: a mechanistic compendium of programmed cell death pathways. *Comput. Struct. Biotechnol. J.* 19:4641–57
78. Yang Y, Ju D, Zhang M, Yang G. 2008. Interleukin-6 stimulates lipolysis in porcine adipocytes. *Endocrine* 33(3):261–69
79. Nguyen KD, Qiu Y, Cui X, Goh YP, Mwangi J, et al. 2011. Alternatively activated macrophages produce catecholamines to sustain adaptive thermogenesis. *Nature* 480(7375):104–8
80. Koutsari C, Jensen MD. 2006. Thematic review series: patient-oriented research. Free fatty acid metabolism in human obesity. *J. Lipid Res.* 47(8):1643–50
81. Flaherty SE, Grijalva A, Xu X, Ables E, Nomani A, Ferrante AW. 2019. A lipase-independent pathway of lipid release and immune modulation by adipocytes. *Science* 363(6430):989–93
82. Jaitin DA, Adlung L, Thaiss CA, Weiner A, Li B, et al. 2019. Lipid-associated macrophages control metabolic homeostasis in a Trem2-dependent manner. *Cell* 178(3):686–98.e14

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

83. Akter SH, Häussler S, Dänicke S, Müller U, von Soosten D, et al. 2011. Physiological and conjugated linoleic acid-induced changes of adipocyte size in different fat depots of dairy cows during early lactation. *J. Dairy Sci.* 94(6):2871–82
84. De Koster J, Strieder-Barboza C, de Souza J, Lock AL, Contreras GA. 2018. Short communication: effects of body fat mobilization on macrophage infiltration in adipose tissue of early lactation dairy cows. *J. Dairy Sci.* 101(8):7608–13
85. Anderson EK, Gutierrez DA, Kennedy A, Hasty AH. 2013. Weight cycling increases T-cell accumulation in adipose tissue and impairs systemic glucose tolerance. *Diabetes* 62(9):3180–88
86. Cottam MA, Caslin HL, Winn NC, Hasty AH. 2022. Multiomics reveals persistence of obesity-associated immune cell phenotypes in adipose tissue during weight loss and weight regain in mice. *Nat. Commun.* 13:2950
87. Zamarron BF, Porsche CE, Luan D, Lucas HR, Mergian TA, et al. 2020. Weight regain in formerly obese mice hastens development of hepatic steatosis due to impaired adipose tissue function. *Obesity* 28(6):1086–97
88. Nam GE, Kim W, Han K, Lee C, Kwon Y, et al. 2020. Body weight variability and the risk of cardiovascular outcomes and mortality in patients with type 2 diabetes: a nationwide cohort study. *Diabetes Care* 43(9):2234–41
89. Zachut M, Moallem U. 2017. Consistent magnitude of postpartum body weight loss within cows across lactations and the relation to reproductive performance. *J. Dairy Sci.* 100(4):3143–54
90. Dechow CD, Rogers GW, Clay JS. 2002. Heritability and correlations among body condition score loss, body condition score, production and reproductive performance. *J. Dairy Sci.* 85(11):3062–70
91. Mann S. 2022. Symposium review: the role of adipose tissue in transition dairy cows: current knowledge and future opportunities. *J. Dairy Sci.* 105(4):3687–701
92. Frigo E, Dechow CD, Pedron O, Cassell BG. 2010. The genetic relationship of body weight and early-lactation health disorders in two experimental herds. *J. Dairy Sci.* 93(3):1184–92
93. van Straten M, Friger M, Shpigel NY. 2009. Events of elevated somatic cell counts in high-producing dairy cows are associated with daily body weight loss in early lactation. *J. Dairy Sci.* 92(9):4386–94
94. Fricke PM, Wiltbank MC, Pursley JR. 2023. The high fertility cycle. *JDS Commun.* 4(2):127–31
95. Vanacker N, Hooper HB, Blouin R, Lacasse P. 2023. Effect of intravenous lipid infusion on biomarkers of insulin resistance and immune functions of dry and nonpregnant dairy cows. *J. Dairy Sci.* 106(3):2113–23
96. Swartz TH, Bradford BJ, Mamedova LK. 2021. Connecting metabolism to mastitis: Hyperketonemia impaired mammary gland defenses during a *Streptococcus uberis* challenge in dairy cattle. *Front. Immunol.* 12:700278
97. Karlsson EA, Sheridan PA, Beck MA. 2010. Diet-induced obesity impairs the T cell memory response to influenza virus infection. *J. Immunol.* 184(6):3127–33
98. Xu Y-X, Wang B, Jing J-N, Ma R, Luo Y-H, et al. 2023. Whole-body adipose tissue multi-omic analyses in sheep reveal molecular mechanisms underlying local adaptation to extreme environments. *Commun. Biol.* 6:159
99. Trim WV, Walhin J-P, Koumanov F, Bouloumié A, Lindsay MA, et al. 2022. Divergent immunometabolic changes in adipose tissue and skeletal muscle with ageing in healthy humans. *J. Physiol.* 600(4):921–47
100. Olagaray KE, Overton MW, Bradford BJ. 2020. Do biological and management reasons for a short or long dry period induce the same effects on dairy cattle productivity? *J. Dairy Sci.* 103(12):11857–75
101. Xue YF, Guo CZ, Hu F, Sun DM, Liu JH, Mao SY. 2019. Molecular mechanisms of lipid metabolism disorder in livers of ewes with pregnancy toxemia. *Animal* 13(5):992–99
102. Gordon JL, LeBlanc SJ, Duffield TF. 2013. Ketosis treatment in lactating dairy cattle. *Vet. Clin. N. Am. Food Anim. Pract.* 29(2):433–45
103. Liu G, Jia H, Ma L, Xu Q, Li XX, et al. 2019. Adipose tissue proteomic analysis in ketotic or healthy Holstein cows in early lactation. *J. Anim. Sci.* 97(7):2837–49
104. Putman AK, Brown JL, Gandy JC, Abuelo A, Sordillo LM. 2019. Oxylipid profiles of dairy cattle vary throughout the transition into early mammary gland involution. *J. Dairy Sci.* 102(3):2481–91

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

105. Putman AK, Brown JL, Gandy JC, Wisnieski L, Sordillo LM. 2018. Changes in biomarkers of nutrient metabolism, inflammation, and oxidative stress in dairy cows during the transition into the early dry period. *J. Dairy Sci.* 101(10):9350–59
106. Richter SH, Kästner N, Loddenkemper D-H, Kaiser S, Sachser N. 2016. A time to wean? Impact of weaning age on anxiety-like behaviour and stability of behavioural traits in full adulthood. *PLOS ONE* 11(12):e0167652
107. Deters EL, Hansen SL. 2020. Invited review: linking road transportation with oxidative stress in cattle and other species. *Appl. Anim. Sci.* 36(2):183–200
108. Dornbach CW, Beenken-Bobb AM, Shike DW, Hansen SL, McCann JC. 2023. Effects of injectable vitamin E before or after transit on receiving phase growth performance, health, and blood parameters of beef steers. *J. Anim. Sci.* 101:skac333
109. Meléndez DM, Marti S, Haley DB, Schwinghamer TD, Schwartzkopf-Genswein KS. 2020. Effect of transport and rest stop duration on the welfare of conditioned cattle transported by road. *PLOS ONE* 15(3):e0228492
110. Meléndez DM, Marti S, Haley DB, Schwinghamer TD, Yang X, Schwartzkopf-Genswein KS. 2022. Effect of rest, post-rest transport duration, and conditioning on performance, behavioural, and physiological welfare indicators of beef calves. *PLOS ONE* 17(12):e0278768
111. Dirandeh E, Ghorbanalinia M, Rezaei-Roodbari A, Colazo MG. 2020. Relationship between body condition score loss and mRNA of genes related to fatty acid metabolism and the endocannabinoid system in adipose tissue of periparturient cows. *Animal* 14(8):1724–32
112. Gordon CJ. 2012. Thermal physiology of laboratory mice: defining thermoneutrality. *J. Therm. Biol.* 37(8):654–85
113. Vialard F, Olivier M. 2020. Thermoneutrality and immunity: How does cold stress affect disease? *Front. Immunol.* 11:588387
114. Abeyta MA, Al-Qaisi M, Horst EA, Mayorga EJ, Rodriguez-Jimenez S, et al. 2023. Effects of dietary antioxidant supplementation on metabolism and inflammatory biomarkers in heat-stressed dairy cows. *J. Dairy Sci.* 106(2):1441–52
115. Mayorga EJ, Renaudeau D, Ramirez BC, Ross JW, Baumgard LH. 2019. Heat stress adaptations in pigs. *Anim. Front.* 9(1):54–61
116. Qu H, Yan H, Lu H, Donkin SS, Ajuwon KM. 2016. Heat stress in pigs is accompanied by adipose tissue-specific responses that favor increased triglyceride storage. *J. Anim. Sci.* 94(5):1884–96
117. Qu H, Ajuwon KM. 2018. Adipose tissue-specific responses reveal an important role of lipogenesis during heat stress adaptation in pigs. *J. Anim. Sci.* 96(3):975–89
118. Reith RR, Sieck RL, Grijalva PC, Swanson RM, Fuller AM, et al. 2022. Transcriptome analyses indicate that heat stress-induced inflammation in white adipose tissue and oxidative stress in skeletal muscle is partially moderated by zilpaterol supplementation in beef cattle. *J. Anim. Sci.* 100(3):skac019
119. Zachut M, Kra G, Livshitz L, Portnick Y, Yakoby S, et al. 2017. Seasonal heat stress affects adipose tissue proteome toward enrichment of the Nrf2-mediated oxidative stress response in late-pregnant dairy cows. *J. Proteom.* 158:52–61
120. Hersoug L-G, Møller P, Loft S. 2018. Role of microbiota-derived lipopolysaccharide in adipose tissue inflammation, adipocyte size and pyroptosis during obesity. *Nutr. Res. Rev.* 31(2):153–63
121. Chirivi M, Rendon CJ, Myers MN, Prom CM, Roy S, et al. 2022. Lipopolysaccharide induces lipolysis and insulin resistance in adipose tissue from dairy cows. *J. Dairy Sci.* 105(1):842–55
122. Chandler TL, Westhoff TA, Overton TR, Lock AL, Van Amburgh ME, et al. 2022. Lipopolysaccharide challenge following intravenous amino acid infusion in postpartum dairy cows: I. Production, metabolic, and hormonal responses. *J. Dairy Sci.* 105(5):4593–610
123. Horst EA, Kvidera SK, Dickson MJ, McCarthy CS, Mayorga EJ, et al. 2019. Effects of continuous and increasing lipopolysaccharide infusion on basal and stimulated metabolism in lactating Holstein cows. *J. Dairy Sci.* 102(4):3584–97
124. Cho DS, Schmitt RE, Dasgupta A, Ducharme AM, Doles JD. 2020. Single-cell deconstruction of post-sepsis skeletal muscle and adipose tissue microenvironments. *J. Cachexia Sarcopenia Muscle* 11(5):1351–63
125. Caputa G, Matsushita M, Sanin DE, Kabat AM, Edwards-Hicks J, et al. 2022. Intracellular infection and immune system cues rewire adipocytes to acquire immune function. *Cell Metab.* 34(5):747–60.e6

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12

126. Hall JR, Rouillard KR, Suchyta DJ, Brown MD, Ahonen MJR, Schoenfisc MH. 2020. Mode of nitric oxide delivery affects antibacterial action. *ACS Biomater. Sci. Eng.* 6(1):433–41
127. Damouche A, Lazure T, Avettand-Fènoël V, Huot N, Dejucq-Rainsford N, et al. 2015. Adipose tissue is a neglected viral reservoir and an inflammatory site during chronic HIV and SIV infection. *PLOS Pathog*. 11(9):e1005153
128. Misumi I, Starmer J, Uchimura T, Beck MA, Magnuson T, Whitmire JK. 2019. Obesity expands a distinct population of T cells in adipose tissue and increases vulnerability to infection. *Cell Rep*. 27(2):514–24.e5
129. Nagajyothi F, Desruisseaux MS, Machado FS, Upadhya R, Zhao D, et al. 2012. Response of adipose tissue to early infection with *Trypanosoma cruzi* (Brazil strain). *J. Infect. Dis.* 205(5):830–40
130. Trindade S, Rijo-Ferreira F, Carvalho T, Pinto-Neves D, Guegan F, et al. 2016. *Trypanosoma brucei* parasites occupy and functionally adapt to the adipose tissue in mice. *Cell Host Microbe* 19(6):837–48
131. Hajri T, Gharib M, Kaul S, Karpeh MSJ. 2017. Association between adipokines and critical illness outcomes. *J. Trauma Acute Care Surg.* 83(3):507–19
132. Ebihara T, Matsumoto H, Matsubara T, Matsuura H, Hirose T, et al. 2021. Adipocytokine profile reveals resistin forming a prognostic-related cytokine network in the acute phase of sepsis. *Shock* 56(5):718–26
133. Kistner TM, Pedersen BK, Lieberman DE. 2022. Interleukin 6 as an energy allocator in muscle tissue. *Nat. Metab.* 4(2):170–79
134. Sierżant K, Perruchot M-H, Merlot E, Le Floc'h N, Gondret F. 2019. Tissue-specific responses of antioxidant pathways to poor hygiene conditions in growing pigs divergently selected for feed efficiency. *BMC Vet. Res.* 15:341
135. Shulman GI. 2014. Ectopic fat in insulin resistance, dyslipidemia, and cardiometabolic disease. *N. Engl. J. Med.* 371(12):1131–41
136. Bobe G, Young JW, Beitz DC. 2004. Invited review: pathology, etiology, prevention, and treatment of fatty liver in dairy cows. *J. Dairy Sci.* 87(10):3105–24
137. Crociati M, Di Giacinto F, Manuali E, Stradaioli G, Sylla L, et al. 2018. Systemic profiling of ectopic fat deposits in the reproductive tract of dairy cows. *Theriogenology* 114:46–53
138. Hausman GJ, Basu U, Du M, Fernyhough-Culver M, Dodson MV. 2014. Intermuscular and intramuscular adipose tissues: bad versus good adipose tissues. *Adipocyte* 3(4):242–55

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:18:12